Field Hours

CSCE 648 Computer Aided Sculpting

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Eight heads and a width

You start a sculpture with proportions you can measure, not with a smooth surface. A standing figure in this exercise is 8 heads tall. The midpoint, at 4 heads, is the pelvis, not the navel and not the knee. You mark those heights on a vertical line before you attach a volume. This course is a composite: the height marks come from figure practice, and the surfaces you hang on them come from the mathematics in the later lectures.

You measure width in the same head unit. If the head is 1 unit wide, a shoulder span of 2 units is narrow here, and a span of 3 units is the width you want. You lay those 3 units on the shoulder line at 6.5 heads up from the ground, above the midpoint. You correct the width before any smooth pass. A later surface will not repair a shoulder that is still 2 units wide.

Heights first, then a width. Smoothing does not repair a short measurement.

8 heads, ground to crown

Midpoint at the pelvis

Shoulders at 6.5 heads

Width of 3 head units

Checking a blocked figure

1. The figure is 8 units tall. One unit is the head height you are using as the ruler.

2. The pelvis mark is at 4. The shoulder line is at 6.5.

3. The shoulders measure 2 units across. You widen them to 3 before you build a surface.

Problem 1. A figure is 8 heads tall. At what height is the pelvis in this exercise?

Answer. 4 heads. That is the midpoint, and the midpoint is the pelvis, not the navel and not the knee.

Problem 2. The shoulders are 2 head units wide. What do you change before smoothing?

Answer. Widen them to 3 head units. A smooth surface will not add the missing unit of width.

Lecture 2 · 90 minutes

A line of action

You pose the figure with a line of action, one curve the spine follows. In the standing rest pose that line is vertical, from the pelvis at height 4 up through the neck. In a lean, the line tilts, and you draw the tilt before you move the rib volume. A lean of 10 degrees carries the head off the vertical through the pelvis. The feet have to

move, or the figure reads as falling rather than leaning.

You keep joint angles inside a named range. A knee here bends from 0 degrees, straight, to 140 degrees, fully folded. You do not bend it to 180 degrees, which would send the shin through the thigh. The shoulder in this exercise swings about 90 degrees forward. You store the pose as those angles on the line of action. Dragging a surface point and hoping the joint follows will twist the volume.

The line of action is the pose. The angles are limits on that line.

Rest pose

Line of action vertical

Head over the pelvis

Knee at 0 degrees

Lean of 10 degrees

Line tilted 10 degrees

Head off the vertical

Feet must move

A knee inside its range

1. The rest knee is 0 degrees. The line of action is vertical through the pelvis at height 4.

2. You set the knee to 140 degrees. The shin folds and does not pass through the thigh.

3. A value of 180 degrees is outside the range. You reject it before you move any surface point.

Problem 1. The line of action tilts 10 degrees and the feet stay under the old vertical. What does the pose read as?

Answer. Falling. The head has left the vertical through the support, and the feet have not moved with the lean.

Problem 2. What knee angles are allowed in this exercise?

Answer. From 0 degrees, straight, through 140 degrees, fully folded. You reject 180 degrees because the shin would pass through the thigh.

Lecture 3 · 90 minutes

A cubic on the limb

You build a silhouette with a few cubics rather than a cloud of points. A limb in side view uses 4 control points: the shoulder, the muscle bulge, the narrow part near the joint, and the next joint. You evaluate with de Casteljau at $t = 0$, $t = 0.5$, and $t = 1$. The curve passes through the first and last controls. You pin those to the joints so the pose angles remain in charge of the endpoints.

You mirror the left and right cubics only in the rest pose. After the 10 degree lean, the two silhouettes are different curves. A mirrored pair would put the same bulge on the compressed side and the stretched side. You

move the middle two controls of the compressed cubic inward by 0.5 head units, and you leave both joint pins fixed. The endpoints stay on the angles you stored.

Joint pins own the endpoints. A lean edits the middle controls on one side only.

1. Pin the joints

2. Place two middle controls

3. Evaluate three parameters

4. Do not mirror a lean

One limb cubic

1. The four controls are joint A, a bulge, a narrow point, and joint B. The curve passes through A and B.

2. At t = 0 you are at A. At t = 1 you are at B. At t = 0.5 the middle controls pull the point off the chord.

3. After a 10 degree lean you shift the compressed side middle controls by 0.5 and you do not copy them across.

Problem 1. Which of the 4 controls must the limb curve pass through?

Answer. The first and the last, the two joint pins. The middle two pull the bulge and the narrow part, and the curve need not pass through them.

Problem 2. Why do you stop mirroring controls after the line of action tilts 10 degrees?

Answer. The compressed side and the stretched side differ. A mirror would give both the same bulge. You edit the compressed cubic by 0.5 head units and keep the joint pins.

Lecture 4 · 90 minutes

Sections along the torso

You turn the curves into a volume by sweeping a cross section along the line of action. The torso here uses an ellipse with radii 1.5 and 1, in head units, swept from the pelvis at height 4 to the shoulder line at height 6.5. That span is 2.5 units. You place 5 cross sections along it so a twist is visible. The sweep is the mechanism. The 5 sections are how you inspect it.

You orient each section with a frame, not by eye. The frame long axis follows the line of action, and the second axis points to the figure left. When the knee bends to 140 degrees, the frames along the shin rotate with the bone, and the sections rotate with the frames. If you leave the sections in the rest orientation, the volume stays vertical while the bone has folded by that full angle.

The line of action carries the frames. The sections inherit those frames.

Line of action -> Five cross sections -> A frame on each -> The swept volume

A torso sweep

1. The sweep runs from height 4 to height 6.5, a span of 2.5 head units.

2. Each ellipse has radii 1.5 and 1. You place 5 of them along the span.

3. Each section takes its orientation from a frame on the line. A bend of the line rotates the frames and the sections together.

Problem 1. How tall is the torso sweep from the pelvis to the shoulder line?

Answer. 2.5 head units. The pelvis is at 4 and the shoulders are at 6.5.

Problem 2. The shin bends to 140 degrees and the cross sections stay in the rest orientation. What is wrong?

Answer. The volume and the bone disagree by 140 degrees. The sections have to rotate with the frames on the line of action.

Lecture 5 · 90 minutes

A cylinder cuts a socket

You cut a volume with constructive solid geometry when the cut is a known solid. The torso is solid A. A cylinder of radius 0.4 is solid B, set where the arm socket belongs. Difference keeps points inside A and outside B. You classify a sample with both inside tests. This is the mathematical half of the composite: the same Boolean you would use on a bracket, applied to a figure.

You place the cylinder from the pose, not from a picture. Its axis leaves the shoulder line at height 6.5 and runs outward. A point 0.2 units from that axis is inside the radius of 0.4, so difference removes it. A point 1 unit from the axis is outside the cylinder, so it stays if it is also inside the torso. After a 10 degree lean you move the cylinder with the shoulder, or you cut the old pose.

Difference removes the socket and leaves torso points outside the cylinder.

Inside the cylinder

Distance 0.2, radius 0.4

Removed by difference

This becomes the socket

Outside the cylinder

Distance 1, radius 0.4

Kept if inside torso

Not part of the cut

Cutting the socket

1. The cylinder has radius 0.4 and its axis runs outward from height 6.5.

2. A point at distance 0.2 is inside the cylinder and is removed from the torso.

3. A point at distance 1 is outside the cylinder. Difference keeps it when it is inside the torso.

Problem 1. A point is inside the torso and inside the socket cylinder. Does difference keep it?

Answer. No. Difference keeps points inside the first solid and outside the second. This point is inside the cylinder, so the cut removes it.

Problem 2. After a 10 degree lean, what do you do with the socket cylinder?

Answer. Move it with the shoulder. The axis was defined on the shoulder line at height 6.5, and that line moved. The old cylinder would cut the rest pose.

Lecture 6 · 90 minutes

A field that bridges

You blend two volumes with an implicit field when a hard Boolean edge is the wrong join. Each ball contributes a field of 1 divided by distance from its center. You sum the fields and keep the surface where the sum equals 1. One ball then has radius 1. A second ball adds field, so the level set bulges toward it and can bridge the gap.

You put the centers on the figure. One sits on the shoulder at height 6.5 and one sits 2 units down the upper arm. If the summed field in the gap reaches 1, the surface connects the balls. You keep level 1 on purpose. A level of 2 shrinks a lone ball to radius 0.5, since the field equals 2 only there. Extract the level after the centers match the pose.

Sum the fields, then extract one level. The level chooses the size.

1. Field is 1 over distance

2. Sum the ball fields

3. Extract level 1

4. A bridge if the sum holds

Two balls along the arm

1. Ball A is centered at the shoulder. Alone, its level set of 1 is a sphere of radius 1.

2. Ball B is centered 2 units down the arm. Alone, it is another sphere of radius 1.

3. In the gap the fields add. If the sum reaches 1, the surface bridges. A level of 2 would shrink a lone ball to radius 0.5.

Problem 1. One ball uses the field 1 over distance, and you extract level 1. What is the radius?

Answer. 1. The field equals 1 when the distance is 1.

Problem 2. You raise the level from 1 to 2 on a single ball. What is the new radius?

Answer. 0.5. The field equals 2 when the distance is 0.5, so the surface shrinks.

Lecture 7 · 90 minutes

From 200 triangles to 800

You refine the mesh only after proportion and pose are right. A coarse figure might use 200 triangles. Splitting each triangle into 4 gives 800. That pass does not move the line of action, does not change the 140 degree knee limit, and does not move the socket you cut with the cylinder. You subdivide so a brush has smaller flats to push. If the pelvis is at the wrong height, 800 triangles only record the mistake more smoothly.

You push a vertex along its normal by 0.1 head units, and you leave the joint pins fixed. A stroke across the muscle moves the region the middle Bezier controls owned. You do not pull a pin at the knee, or the bend stops matching the stored angle. After the stroke you look at the silhouette again. If the 0.1 step broke the bulge you wanted, you undo it rather than smoothing the error into the neighbors.

Density comes last. A brush step does not get to move a joint pin.

Fix the pose first

200 triangles become 800

Push 0.1 along the normal

Joint pins stay fixed

One brush step

1. The mesh has 200 triangles. You subdivide each into 4, getting 800, only after the pelvis is at height 4.

2. A stroke moves one non-pin vertex 0.1 units along its normal.

3. The knee pin does not move, so the stored angle, up to the 140 degree limit, still matches the mesh.

Problem 1. You split each of 200 triangles into 4. How many triangles result, and does that fix a low pelvis?

Answer. 800 triangles. The split does not move the pelvis. A height error remains, only on a finer mesh.

Problem 2. A stroke would move the knee pin by 0.1. What do you do?

Answer. Leave the pin fixed. Moving it would pull the joint off the stored angle. A non-pin vertex may move 0.1 along its normal.

Lecture 8 · 90 minutes

Review in order

You can now block a figure at 8 heads, set a knee from 0 to 140 degrees, build a limb with a 4-point cubic, sweep a torso with 5 sections over 2.5 units, cut a socket by difference, and bridge an arm where a field equals 1. You can then go from 200 triangles to 800 and push a vertex by 0.1 without moving a pin. Figure practice and the surface math stay on one model.

You review in that order because a later tool hides an earlier error. An implicit bridge at level 1 will paper over a shoulder that is still 2 units wide instead of 3. A Boolean socket left on the rest pose misses the shoulder after a 10 degree lean. You check heights, then angles, then the curve pins, then the field level, and only then the triangle count. The composite is the order, not a pile of tools.

Later tools hide earlier mistakes, so the check follows the build order.

Heights and angles -> Curves and volumes -> Boolean or field -> Then subdivide

A leaned figure, checked in order

1. The pelvis is at 4, the shoulders are at 6.5 and 3 units wide, and the line of action tilts 10 degrees.

2. The knee is at 140, not 180. The limb cubics are not mirrored. The socket cylinder moved with the shoulder.

3. The arm bridge is the level set at 1. Only then do you go from 200 triangles to 800.

Problem 1. Combine the lean and the implicit surface. The line of action leans 10 degrees after you placed two ball centers. What must move before you extract level 1?

Answer. The ball centers. They sit on the shoulder at height 6.5 and 2 units down the arm, and those landmarks moved with the lean. Level 1 on the old centers would bridge the rest pose.

Problem 2. Combine the sweep and the brush. Five sections run from height 4 to 6.5, then a stroke moves a vertex 0.1 along its normal, including a joint pin. Which part do you keep?

Answer. Only the move on non-pin vertices. The pin stays, so the sweep joint and the stored angle still agree. The 0.1 step is allowed on the rest of the section.